Gowtham S

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PROFESSIONAL SUMMARY:

- 9+ Years of embedded software development experience spanning from designing, coding, and unit-testing of applications in automotive domain
- Experience in developing Embedded based Applications involving C
- Experience in developing Embedded based Bootloader software involving C for Application Flash.
- Extensive experience in AUTOSAR (COM, DCM, DEM, and Software Component), integration and Testing.
- Experience in using AUTOSAR configuration and generation tools like Vector Davinci
- Experience in Debugging Skills and using tools like S32 Design Studio and other tools.
- Experience in Linux basic knowledge.
- Working on Implementing functionalities (State Machine for the Application Layer).
- DID and PID implementation using Vector Tools
- Contrived the requirement documents from the scratch from the VOC of customers/clients
- Project Estimation comprises the due diligence report of the existing product.

Technical Skills

Operating Systems	Linux and Windows
Programming Languages	Embedded C
Scripting Languages	CAPL
Protocols	CAN, UDS, ISO 15765-2, ISO 26262-Part6, LIN, KWP over UART
Tools	GCC, make, SmartGit repository, Candela Studio, CANalyzer, CANoe, Code warrior, Cosmic, VectorGeny, Davinci Configurator, Davinci Developer, Vector Vflash, Canoe Diva, S32K Design Studio, NXP Open Bus Configurator, DevSuite, Winidea, AEE-Pro , Re-programming Tools and DigiView.
Debuggers, Controllers and Processor	GDB, True-time simulator & Real-time Debugger, ZAP Debugger, S32K146 Controller, S12 controller, SAMA5D29

Experience Summary:

- Company: Capgemini
- Role: Senior Professional
- Duration: Dec 2016 to Present

Educational Details:

- B. Tech from Karpagam College of Engineering, Coimbatore in 2016

Project Details:

Project 1:
Title: Diagnostic Tool- Firmware Development
Tools: CANalyzer/Canoe, DigiView – Logic Analyzer, Peak and Minicom.
Programming Language: Embedded C

Description: Working on Firmware development for the Diagnostic tool to support different protocols like CAN, CANFD, DOIP and KWP in Linux Platform

Roles and Responsibilities:

- Project handling: Interactions with OEMs
- Handled USB packet, converted into corresponding Protocol format.
- Implementation of KWP 2000 over UART in Linux Platform: Includes Initialization, Communication, timers and corruption handling.
- Implementation of CAN protocol in Linux platform: Configured Socket for CAN and handled communication.
- Implementation of CANFD protocol in Linux platform: Configured Socket for CAN and handled communication.
- Tested CAN, DOIP, KWP and LIN protocol in OEM tools.

Project 2:
Title: Functional Safety- Product Development at Software Level
Tools: NA
Programming Language: NA

Description: Preparing checklist for Product development at software level to ensure the compliance of Functional Safety (ISO 26262- Part 6).

Roles and Responsibilities:

- Prepared work products like Safety plan, FUSA V&V test Strategy and Safety case as per Part 6 of ISO26262.
- Performed DFMEA for Audio and GPU drivers.
- Prepared Functional safety requirement for Audio and GPU.
- Prepared ASIL compliance review document.

Project 3:
Title: Bootloader Development
Tools: Winidea, AEE-Pro, CANalyzer/Canoe, Reprogramming tools
Programming Language: Embedded C

Description: This software is used to re-programming the Application and Calibration data for new update of Functionality. Also, this software will validate the Valid Application and necessary check to make sure the Software is in well-Shaped.
Roles and Responsibilities:

- Bootloader Jump logic implementation for jumping to valid application block
- DID Configuration and Implementation.
- Generated COM module (Part of Bootloader) using AEEE tool. Configured and generated module code. Integrated the generated code with main code for Bootloader
- Re-programming sequence implementation including all necessary services.
- Re-programming testing

Project 4:
Title: Power Step System Module
Tools: NXP S32 Design Studio, VectorGeny, NXP Open Bus and CANalyzer/CANoe
Programming Language: Embedded C

Description: This Module comprises ARB feature; which is used in higher end SUVs, to assist user with stepper - board retraction/deployment; whenever door closes/open due to the higher ground clearance of SUVs.
Roles and Responsibilities:

- Project handling along with customer SPOC. Interactions with OEMs
- Handling of variants through Configurations for Different Programs of same OEM
- State Machine implementation for the Application Layer
- DIDs and PID implementation
- Code Integration: Vector GENy is used to configure and generate LIN stack layers. Integrated the code with application layers for full functionality
- MISRA Compliance: Ensured MISRA compliance using Understand-C tool
- Unit testing: Created unit test cases and tested
- Bug Fixing: Supported customer changes, bugs etc, involved in reviews and impact analysis
- DID implementation / testing on bootloader.
- Panel Design: Designed CANoe panels for validating the functionality

Project 5:
Title: Running Board Module
Tools: Davinci Developer, Davinci Configurator, Cosmic compiler, ZAP Debugger and CANalyzer/CANoe
Programming Language: Embedded C

Description This Module comprises ARB feature; which is used in higher end SUVs, to assist user with stepper - board retraction/deployment; whenever door closes/open due to the higher ground clearance of SUVs.

Roles and Responsibilities:

- Project handling along with customer SPOC. Interactions with OEMs
- Code Integration: Vector Davinci is used to configure and generate LIN stack layers. Integrated the code with application layers for full functionality
- Application State Machine implementation
- Configured and tested Candela Diagnostic Description (CDD) file.
- DIDs and PIDs configuration using CDD file in DCM module.
- Configured and generated COM module using Davinci Configurator.
- Software Component (SWC) Implemented for COM module using Davinci Developer
- MISRA Compliance: Ensured MISRA compliance using Understand-C tool
- Unit testing: Created unit test cases and tested

Project 6:
Title: Illumination Power Running Board Module
Tools: VectorGeny, CodeWarrior compiler, Real-time Debugger and CANalyzer/CANoe
Programming Language: Embedded C

Description: This Module comprises ARB feature; which is used in higher end SUVs, to assist user with stepper - board retraction/deployment; whenever door closes/open due to the higher ground clearance of SUVs.

Roles and Responsibilities:

- Candela Diagnostic Description file configuration for the Module.
- Preformed Functionality testing using Simulation tools (CANoe)
- Performed Diagnostic testing using OEM Tools
- MISRA Compliance: Ensured MISRA compliance using Understand-C tool
- Unit testing: Created unit test cases and tested
- Bug Fixing: Supported customer changes, bugs etc, involved in reviews and impact analysis
- CAPL and Panel Design: Designed CANoe panels for validating the functionality